

## A.8 GOOD DATASET ADDITIONAL RESULTS

We also include results on additional node classification benchmarks featuring distribution shift in Table 12. In Table 13, we compare models without G- $\Delta$ UQ to the use of G- $\Delta$ UQ with randomly sampled anchors at the first or second layer.

**Table 12: Additional Node Classification Benchmarks.** Here, we compare accuracy and calibration error of G- $\Delta$ UQ and "no G- $\Delta$ UQ" (vanilla) models on 4 node classification benchmarks across concept and covariate shifts. First, we note that across all our evaluations, without any posthoc calibration, G- $\Delta$ UQ is superior to the vanilla model on nearly every benchmark for better or same accuracy (8/8 benchmarks) and better calibration error (7/8), often with a significant gain in calibration performance. However, due to the challenging nature of these shifts, achieving state-of-the-art calibration performance often requires the use of post-hoc calibration methods – so we also evaluate how these posthoc methods can be elevated when combined with G- $\Delta$ UQ (versus the vanilla variant). When combined with popular posthoc methods, we highlight that performance improves across the board, when combined with G- $\Delta$ UQ (including in WebKB and CBAS-Concept). For example, on WebKB – across the 9 calibration methods considered, “G- $\Delta$ UQ + calibration method” improves or maintains the calibration performance of the analogous “no G- $\Delta$ UQ + calibration method” in 7/9 (concept) and 6/9 (covariate). In CBAS, calibration is improved or maintained as the no-G- $\Delta$ UQ version on 5/9 (concept) and 9/9 (covariate). In all cases, this is achieved with little or no compromise on classification accuracy (often improving over “no G- $\Delta$ UQ” variant). We also emphasize that, across all the 8 evaluation sets (4 datasets x 2 shift types) in Table 10, the best performance is almost always obtained with a GDUQ variant: (accuracy: 8/8) as well as best calibration (6/8) or second best (2/8).

| Dataset | Domain     | Calibration    | Shift: Concept          |                                   |                                   |                                   | Shift: Covariate                  |                                   |                                   |                                   |
|---------|------------|----------------|-------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
|         |            |                | Accuracy ( $\uparrow$ ) |                                   | ECE ( $\downarrow$ )              |                                   | Accuracy ( $\uparrow$ )           |                                   | ECE ( $\downarrow$ )              |                                   |
|         |            |                | No G- $\Delta$ UQ       | G- $\Delta$ UQ                    | No G- $\Delta$ UQ                 | G- $\Delta$ UQ                    | No G- $\Delta$ UQ                 | G- $\Delta$ UQ                    | No G- $\Delta$ UQ                 | G- $\Delta$ UQ                    |
| WebKB   | University | X              | 0.253 $\pm$ 0.003       | <b>0.281<math>\pm</math>0.009</b> | 0.67 $\pm$ 0.061                  | 0.593 $\pm$ 0.025                 | 0.122 $\pm$ 0.029                 | 0.115 $\pm$ 0.041                 | 0.599 $\pm$ 0.091                 | 0.525 $\pm$ 0.033                 |
|         |            | CAGCN          | 0.253 $\pm$ 0.005       | 0.268 $\pm$ 0.008                 | 0.452 $\pm$ 0.14                  | 0.473 $\pm$ 0.12                  | 0.122 $\pm$ 0.018                 | 0.092 $\pm$ 0.161                 | 0.355 $\pm$ 0.227                 | 0.396 $\pm$ 0.161                 |
|         |            | Dirichlet      | 0.229 $\pm$ 0.018       | 0.22 $\pm$ 0.022                  | 0.472 $\pm$ 0.06                  | 0.472 $\pm$ 0.03                  | <b>0.244<math>\pm</math>0.105</b> | <b>0.295<math>\pm</math>0.044</b> | <b>0.299<math>\pm</math>0.092</b> | <b>0.328<math>\pm</math>0.044</b> |
|         |            | ETS            | 0.253 $\pm$ 0.005       | 0.273 $\pm$ 0.012                 | 0.64 $\pm$ 0.06                   | 0.575 $\pm$ 0.019                 | 0.121 $\pm$ 0.021                 | 0.084 $\pm$ 0.027                 | 0.539 $\pm$ 0.112                 | 0.499 $\pm$ 0.027                 |
|         |            | GATS           | 0.253 $\pm$ 0.005       | 0.273 $\pm$ 0.01                  | 0.608 $\pm$ 0.008                 | 0.485 $\pm$ 0.02                  | 0.122 $\pm$ 0.018                 | 0.079 $\pm$ 0.029                 | 0.455 $\pm$ 0.057                 | 0.376 $\pm$ 0.029                 |
|         |            | IRM            | 0.251 $\pm$ 0.005       | 0.266 $\pm$ 0.011                 | <b>0.342<math>\pm</math>0.017</b> | <b>0.349<math>\pm</math>0.006</b> | 0.097 $\pm$ 0.04                  | 0.046 $\pm$ 0.013                 | 0.352 $\pm$ 0.037                 | 0.422 $\pm$ 0.013                 |
|         |            | Orderinvariant | 0.253 $\pm$ 0.005       | 0.27 $\pm$ 0.01                   | 0.628 $\pm$ 0.026                 | 0.564 $\pm$ 0.024                 | 0.122 $\pm$ 0.018                 | 0.106 $\pm$ 0.065                 | 0.545 $\pm$ 0.079                 | 0.47 $\pm$ 0.065                  |
|         |            | Spline         | 0.237 $\pm$ 0.012       | 0.257 $\pm$ 0.023                 | 0.436 $\pm$ 0.029                 | 0.386 $\pm$ 0.034                 | 0.122 $\pm$ 0.013                 | 0.171 $\pm$ 0.056                 | 0.472 $\pm$ 0.031                 | 0.39 $\pm$ 0.056                  |
|         |            | VS             | 0.253 $\pm$ 0.005       | <b>0.275<math>\pm</math>0.011</b> | 0.67 $\pm$ 0.009                  | 0.588 $\pm$ 0.011                 | 0.122 $\pm$ 0.018                 | 0.095 $\pm$ 0.014                 | 0.602 $\pm$ 0.044                 | 0.507 $\pm$ 0.014                 |
| Cora    | Degree     | X              | 0.581 $\pm$ 0.003       | 0.595 $\pm$ 0.003                 | 0.307 $\pm$ 0.009                 | 0.13 $\pm$ 0.011                  | 0.47 $\pm$ 0.002                  | 0.518 $\pm$ 0.014                 | 0.348 $\pm$ 0.032                 | 0.141 $\pm$ 0.008                 |
|         |            | CAGCN          | 0.581 $\pm$ 0.003       | <b>0.597<math>\pm</math>0.002</b> | 0.135 $\pm$ 0.009                 | 0.128 $\pm$ 0.025                 | 0.47 $\pm$ 0.002                  | <b>0.522<math>\pm</math>0.025</b> | 0.256 $\pm$ 0.08                  | 0.231 $\pm$ 0.025                 |
|         |            | Dirichlet      | 0.534 $\pm$ 0.007       | 0.551 $\pm$ 0.004                 | 0.12 $\pm$ 0.004                  | 0.196 $\pm$ 0.003                 | 0.414 $\pm$ 0.007                 | 0.449 $\pm$ 0.01                  | 0.163 $\pm$ 0.002                 | 0.356 $\pm$ 0.01                  |
|         |            | ETS            | 0.581 $\pm$ 0.003       | <b>0.596<math>\pm</math>0.004</b> | 0.301 $\pm$ 0.009                 | 0.116 $\pm$ 0.018                 | 0.47 $\pm$ 0.002                  | <b>0.523<math>\pm</math>0.003</b> | 0.31 $\pm$ 0.077                  | 0.141 $\pm$ 0.003                 |
|         |            | GATS           | 0.581 $\pm$ 0.003       | <b>0.596<math>\pm</math>0.004</b> | 0.185 $\pm$ 0.018                 | 0.229 $\pm$ 0.039                 | 0.47 $\pm$ 0.002                  | 0.521 $\pm$ 0.011                 | 0.211 $\pm$ 0.004                 | 0.308 $\pm$ 0.011                 |
|         |            | IRM            | 0.582 $\pm$ 0.002       | 0.597 $\pm$ 0.002                 | 0.125 $\pm$ 0.001                 | 0.102 $\pm$ 0.002                 | 0.469 $\pm$ 0.001                 | <b>0.522<math>\pm</math>0.004</b> | 0.194 $\pm$ 0.005                 | <b>0.13<math>\pm</math>0.004</b>  |
|         |            | Orderinvariant | 0.581 $\pm$ 0.003       | 0.592 $\pm$ 0.002                 | 0.226 $\pm$ 0.024                 | 0.213 $\pm$ 0.049                 | 0.47 $\pm$ 0.002                  | 0.498 $\pm$ 0.027                 | 0.318 $\pm$ 0.042                 | 0.196 $\pm$ 0.027                 |
|         |            | Spline         | 0.571 $\pm$ 0.003       | 0.595 $\pm$ 0.003                 | <b>0.080<math>\pm</math>0.004</b> | <b>0.068<math>\pm</math>0.004</b> | 0.459 $\pm$ 0.003                 | 0.52 $\pm$ 0.004                  | 0.158 $\pm$ 0.01                  | <b>0.098<math>\pm</math>0.004</b> |
|         |            | VS             | 0.581 $\pm$ 0.003       | <b>0.596<math>\pm</math>0.004</b> | 0.306 $\pm$ 0.004                 | 0.127 $\pm$ 0.002                 | 0.47 $\pm$ 0.001                  | <b>0.522<math>\pm</math>0.005</b> | 0.345 $\pm$ 0.005                 | 0.146 $\pm$ 0.005                 |
| Cora    | Word       | X              | 0.607 $\pm$ 0.003       | 0.628 $\pm$ 0.001                 | 0.284 $\pm$ 0.009                 | 0.111 $\pm$ 0.013                 | 0.603 $\pm$ 0.004                 | 0.633 $\pm$ 0.031                 | 0.263 $\pm$ 0.004                 | 0.118 $\pm$ 0.019                 |
|         |            | CAGCN          | 0.607 $\pm$ 0.002       | 0.628 $\pm$ 0.002                 | 0.138 $\pm$ 0.011                 | 0.236 $\pm$ 0.019                 | 0.603 $\pm$ 0.004                 | 0.634 $\pm$ 0.035                 | 0.129 $\pm$ 0.009                 | 0.253 $\pm$ 0.035                 |
|         |            | Dirichlet      | 0.579 $\pm$ 0.007       | 0.588 $\pm$ 0.006                 | 0.105 $\pm$ 0.011                 | 0.168 $\pm$ 0.005                 | 0.562 $\pm$ 0.007                 | 0.578 $\pm$ 0.007                 | 0.095 $\pm$ 0.006                 | 0.269 $\pm$ 0.007                 |
|         |            | ETS            | 0.607 $\pm$ 0.002       | 0.628 $\pm$ 0.002                 | 0.282 $\pm$ 0.002                 | 0.11 $\pm$ 0.003                  | 0.603 $\pm$ 0.004                 | 0.634 $\pm$ 0.013                 | 0.243 $\pm$ 0.023                 | 0.106 $\pm$ 0.013                 |
|         |            | GATS           | 0.607 $\pm$ 0.002       | 0.628 $\pm$ 0.002                 | 0.166 $\pm$ 0.009                 | 0.261 $\pm$ 0.028                 | 0.603 $\pm$ 0.004                 | <b>0.635<math>\pm</math>0.037</b> | 0.16 $\pm$ 0.015                  | 0.293 $\pm$ 0.037                 |
|         |            | IRM            | 0.608 $\pm$ 0.001       | 0.63 $\pm$ 0.002                  | 0.115 $\pm$ 0.002                 | 0.088 $\pm$ 0.003                 | 0.602 $\pm$ 0.003                 | <b>0.635<math>\pm</math>0.004</b> | 0.106 $\pm$ 0.002                 | 0.098 $\pm$ 0.004                 |
|         |            | Orderinvariant | 0.607 $\pm$ 0.002       | 0.624 $\pm$ 0.002                 | 0.174 $\pm$ 0.024                 | 0.201 $\pm$ 0.061                 | 0.603 $\pm$ 0.004                 | 0.621 $\pm$ 0.076                 | 0.154 $\pm$ 0.022                 | 0.202 $\pm$ 0.076                 |
|         |            | Spline         | 0.598 $\pm$ 0.005       | <b>0.629<math>\pm</math>0.002</b> | <b>0.073<math>\pm</math>0.002</b> | <b>0.062<math>\pm</math>0.005</b> | 0.591 $\pm$ 0.002                 | <b>0.635<math>\pm</math>0.004</b> | <b>0.063<math>\pm</math>0.006</b> | <b>0.053<math>\pm</math>0.004</b> |
|         |            | VS             | 0.607 $\pm$ 0.001       | <b>0.63<math>\pm</math>0.002</b>  | 0.283 $\pm$ 0.003                 | 0.111 $\pm$ 0.003                 | 0.603 $\pm$ 0.004                 | <b>0.636<math>\pm</math>0.003</b> | 0.261 $\pm$ 0.005                 | 0.119 $\pm$ 0.003                 |
| CBAS    | Color      | X              | 0.83 $\pm$ 0.014        | 0.829 $\pm$ 0.011                 | 0.169 $\pm$ 0.013                 | 0.151 $\pm$ 0.014                 | 0.703 $\pm$ 0.015                 | 0.746 $\pm$ 0.027                 | 0.266 $\pm$ 0.02                  | 0.169 $\pm$ 0.018                 |
|         |            | CAGCN          | 0.83 $\pm$ 0.013        | 0.83 $\pm$ 0.013                  | <b>0.137<math>\pm</math>0.011</b> | 0.143 $\pm$ 0.022                 | 0.703 $\pm$ 0.019                 | 0.749 $\pm$ 0.033                 | 0.25 $\pm$ 0.021                  | 0.186 $\pm$ 0.017                 |
|         |            | Dirichlet      | 0.801 $\pm$ 0.02        | 0.806 $\pm$ 0.008                 | 0.161 $\pm$ 0.012                 | 0.17 $\pm$ 0.01                   | 0.671 $\pm$ 0.018                 | 0.771 $\pm$ 0.03                  | 0.241 $\pm$ 0.029                 | 0.217 $\pm$ 0.017                 |
|         |            | ETS            | 0.83 $\pm$ 0.013        | 0.827 $\pm$ 0.014                 | 0.146 $\pm$ 0.013                 | 0.164 $\pm$ 0.007                 | 0.703 $\pm$ 0.019                 | 0.76 $\pm$ 0.037                  | 0.28 $\pm$ 0.023                  | 0.176 $\pm$ 0.019                 |
|         |            | GATS           | 0.83 $\pm$ 0.013        | 0.83 $\pm$ 0.021                  | 0.16 $\pm$ 0.009                  | 0.173 $\pm$ 0.021                 | 0.703 $\pm$ 0.019                 | 0.751 $\pm$ 0.016                 | 0.236 $\pm$ 0.039                 | <b>0.16<math>\pm</math>0.015</b>  |
|         |            | IRM            | 0.829 $\pm$ 0.013       | <b>0.839<math>\pm</math>0.015</b> | 0.142 $\pm$ 0.009                 | <b>0.133<math>\pm</math>0.006</b> | 0.72 $\pm$ 0.019                  | <b>0.803<math>\pm</math>0.04</b>  | 0.207 $\pm$ 0.035                 | <b>0.158<math>\pm</math>0.017</b> |
|         |            | Orderinvariant | 0.83 $\pm$ 0.013        | 0.803 $\pm$ 0.008                 | 0.174 $\pm$ 0.006                 | 0.173 $\pm$ 0.009                 | 0.703 $\pm$ 0.019                 | 0.766 $\pm$ 0.045                 | 0.261 $\pm$ 0.017                 | 0.194 $\pm$ 0.031                 |
|         |            | Spline         | 0.82 $\pm$ 0.016        | 0.824 $\pm$ 0.011                 | 0.159 $\pm$ 0.009                 | 0.16 $\pm$ 0.014                  | 0.683 $\pm$ 0.019                 | 0.786 $\pm$ 0.038                 | 0.225 $\pm$ 0.034                 | 0.179 $\pm$ 0.035                 |
|         |            | VS             | 0.829 $\pm$ 0.012       | <b>0.840<math>\pm</math>0.011</b> | 0.166 $\pm$ 0.011                 | 0.146 $\pm$ 0.012                 | 0.717 $\pm$ 0.019                 | <b>0.809<math>\pm</math>0.008</b> | 0.242 $\pm$ 0.019                 | 0.182 $\pm$ 0.014                 |

Table 13: **Layerwise Anchoring for Node Classification Datasets with Intermediate Representation Distributions.** Here, we provide preliminary results for performing layerwise anchoring when performing node classification. We fit a gaussian distribution over the representations (similar to node feature anchoring) and then sample anchors from this distribution. We fit a gaussian distribution over the representations (similar to node feature anchoring) and then sample anchors from this distribution. We see that these alternative strategies do provide benefits in some cases, but overall, our original input node feature anchoring strategy is more performant.

| Dataset | Domain     | Calibration    | Shift: Concept     |                    |                    |                    |                    |                    | Shift: Covariate |                    |                    |                    |                    |                    |
|---------|------------|----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|         |            |                | Accuracy (↑)       |                    |                    | ECE (↓)            |                    |                    | Accuracy (↑)     |                    |                    | ECE (↓)            |                    |                    |
|         |            |                | No G-Δ UQ          | Random 1           | Random 2           | No G-Δ UQ          | Random 1           | Random 2           | No G-Δ UQ        | Random 1           | Random 2           | No G-Δ UQ          | Random 1           | Random 2           |
| CBAS    | Color      | Dirichlet      | 0.801±0.02         | 0.765±0.012        | 0.839±0.023        | 0.161±0.012        | 0.301±0.018        | 0.234±0.027        | 0.671±0.018      | <b>0.74±0.023</b>  | 0.689±0.032        | <b>0.241±0.029</b> | 0.349±0.04         | 0.381±0.029        |
|         |            | ETS            | 0.83±0.013         | 0.819±0.012        | 0.82±0.088         | <b>0.146±0.013</b> | 0.23±0.017         | 0.257±0.021        | 0.703±0.019      | 0.638±0.051        | 0.686±0.026        | 0.28±0.023         | 0.347±0.037        | 0.334±0.028        |
|         |            | IRM            | 0.829±0.013        | 0.821±0.019        | <b>0.885±0.026</b> | <b>0.142±0.009</b> | 0.219±0.012        | 0.206±0.066        | 0.72±0.019       | 0.617±0.084        | 0.693±0.026        | <b>0.267±0.035</b> | 0.363±0.03         | 0.299±0.036        |
|         |            | Orderinvariant | 0.83±0.013         | 0.813±0.015        | 0.819±0.028        | 0.174±0.006        | 0.255±0.015        | 0.236±0.006        | 0.703±0.019      | 0.831±0.088        | 0.636±0.026        | 0.267±0.035        | 0.286±0.039        | 0.303±0.062        |
|         |            | Spline         | 0.82±0.016         | 0.814±0.022        | 0.839±0.035        | 0.159±0.009        | 0.235±0.017        | 0.196±0.036        | 0.683±0.019      | 0.621±0.052        | <b>0.757±0.026</b> | 0.225±0.034        | 0.312±0.026        | 0.331±0.024        |
| Cora    | Degree     | VS             | 0.829±0.012        | 0.817±0.017        | <b>0.91±0.006</b>  | 0.166±0.011        | 0.251±0.012        | 0.259±0.021        | 0.717±0.019      | 0.593±0.038        | 0.695±0.051        | 0.242±0.019        | 0.38±0.037         | 0.359±0.02         |
|         |            | Dirichlet      | 0.534±0.007        | 0.483±0.014        | 0.423±0.007        | 0.12±0.004         | 0.355±0.004        | 0.347±0.004        | 0.414±0.007      | 0.466±0.073        | 0.425±0.005        | 0.163±0.002        | 0.315±0.042        | 0.345±0.007        |
|         |            | ETS            | <b>0.581±0.003</b> | 0.562±0.01         | 0.496±0.002        | 0.301±0.009        | 0.297±0.009        | 0.289±0.006        | 0.47±0.002       | 0.498±0.119        | 0.34±0.076         | 0.31±0.077         | 0.511±0.005        | 0.329±0.008        |
|         |            | IRM            | <b>0.582±0.002</b> | 0.567±0.011        | 0.492±0.003        | 0.125±0.001        | <b>0.072±0.003</b> | 0.116±0.006        | 0.469±0.001      | 0.499±0.117        | 0.508±0.005        | 0.194±0.005        | <b>0.094±0.009</b> | <b>0.105±0.006</b> |
|         |            | Orderinvariant | <b>0.581±0.003</b> | 0.566±0.004        | 0.495±0.002        | 0.226±0.024        | 0.151±0.015        | 0.14±0.008         | 0.47±0.002       | 0.499±0.107        | 0.108±0.034        | 0.318±0.042        | 0.506±0.005        | 0.093±0.009        |
| Cora    | Word       | Spline         | 0.571±0.003        | 0.561±0.011        | 0.493±0.005        | <b>0.080±0.004</b> | 0.111±0.01         | 0.119±0.005        | 0.459±0.003      | 0.499±0.112        | 0.508±0.006        | 0.158±0.01         | <b>0.105±0.03</b>  | 0.127±0.012        |
|         |            | VS             | <b>0.581±0.003</b> | 0.571±0.002        | 0.279±0.009        | 0.306±0.004        | 0.493±0.008        | 0.272±0.009        | 0.47±0.001       | <b>0.511±0.091</b> | <b>0.51±0.002</b>  | 0.345±0.005        | 0.347±0.051        | 0.323±0.007        |
|         |            | Dirichlet      | 0.579±0.007        | 0.581±0.004        | 0.504±0.004        | 0.105±0.011        | 0.271±0.011        | 0.285±0.002        | 0.562±0.007      | 0.586±0.009        | 0.497±0.01         | 0.095±0.006        | 0.264±0.022        | 0.275±0.007        |
|         |            | ETS            | 0.607±0.002        | <b>0.641±0.003</b> | 0.575±0.003        | 0.282±0.002        | 0.352±0.012        | 0.328±0.007        | 0.603±0.004      | 0.633±0.003        | 0.567±0.004        | 0.243±0.023        | 0.377±0.023        | 0.374±0.006        |
|         |            | IRM            | 0.608±0.001        | <b>0.642±0.002</b> | 0.574±0.003        | 0.115±0.002        | 0.106±0.004        | 0.154±0.005        | 0.602±0.003      | 0.635±0.004        | 0.569±0.003        | 0.106±0.002        | 0.136±0.012        | 0.173±0.007        |
| WebKB   | University | Orderinvariant | 0.607±0.002        | <b>0.642±0.004</b> | 0.573±0.004        | 0.174±0.024        | 0.109±0.011        | 0.107±0.01         | 0.603±0.004      | <b>0.638±0.004</b> | 0.566±0.004        | 0.154±0.022        | 0.087±0.006        | 0.073±0.004        |
|         |            | Spline         | 0.598±0.005        | <b>0.641±0.002</b> | 0.576±0.004        | <b>0.073±0.002</b> | 0.076±0.004        | <b>0.068±0.007</b> | 0.591±0.002      | 0.632±0.002        | 0.568±0.003        | <b>0.063±0.006</b> | <b>0.066±0.005</b> | 0.077±0.004        |
|         |            | VS             | 0.607±0.001        | 0.639±0.003        | 0.583±0.005        | 0.283±0.003        | 0.345±0.007        | 0.335±0.012        | 0.603±0.004      | <b>0.637±0.004</b> | 0.579±0.004        | 0.261±0.005        | 0.396±0.028        | 0.384±0.005        |
|         |            | Dirichlet      | 0.229±0.018        | 0.214±0.000        | 0.228±0.012        | 0.472±0.06         | 0.56±0.000         | 0.552±0.041        | 0.244±0.105      |                    | 0.347±0.012        | 0.299±0.092        |                    | 0.429±0.05         |
|         |            | ETS            | <b>0.253±0.005</b> | <b>0.279±0.000</b> | 0.234±0.01         | 0.64±0.06          | 0.437±0.000        | <b>0.33±0.022</b>  | 0.121±0.021      |                    | 0.225±0.013        | 0.539±0.112        |                    | 0.258±0.028        |

Table 14: **Layerwise Anchoring for Node Classification Datasets with Random Shuffling.** Here, we provide preliminary results for performing layerwise anchoring when performing node classification. We use random shuffling (similar to the proposed hidden layer strategy) to create the intermediate representations. We see that these alternative strategies do provide benefits.

| Dataset | Domain     | Calibration    | Shift: Concept     |                    |             |                    |                    |                    | Shift: Covariate   |                    |                    |                    |                    |                   |
|---------|------------|----------------|--------------------|--------------------|-------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------------|
|         |            |                | Accuracy (↑)       |                    |             | ECE (↓)            |                    |                    | Accuracy (↑)       |                    |                    | ECE (↓)            |                    |                   |
|         |            |                | No G-Δ UQ          | Batch 1            | Batch 2     | No G-Δ UQ          | Batch 1            | Batch 2            | No G-Δ UQ          | Batch 1            | Batch 2            | No G-Δ UQ          | Batch 1            | Batch 2           |
| CBAS    | Color      | Dirichlet      | 0.801±0.02         | 0.757±0.045        | 0.58±0.046  | 0.161±0.012        | 0.309±0.059        | 0.431±0.033        | 0.671±0.018        | 0.548±0.035        | 0.629±0.019        | 0.241±0.029        | 0.48±0.03          | 0.407±0.01        |
|         |            | ETS            | <b>0.83±0.013</b>  | 0.699±0.036        | 0.637±0.014 | <b>0.146±0.013</b> | 0.265±0.013        | 0.258±0.015        | 0.703±0.019        | 0.562±0.087        | 0.507±0            | 0.28±0.023         | 0.37±0.021         | 0.333±0.02        |
|         |            | IRM            | 0.829±0.013        | 0.711±0.031        | 0.724±0.029 | <b>0.142±0.009</b> | 0.284±0.022        | 0.291±0.02         | <b>0.72±0.019</b>  | 0.59±0.079         | 0.657±0.037        | <b>0.267±0.035</b> | 0.336±0.032        | 0.285±0.037       |
|         |            | Orderinvariant | <b>0.83±0.013</b>  | 0.788±0.007        | 0.574±0.051 | 0.174±0.006        | 0.268±0.023        | 0.208±0.055        | 0.703±0.019        | 0.61±0.011         | 0.57±0.019         | 0.261±0.017        | 0.334±0.035        | 0.249±0.037       |
|         |            | Spline         | 0.82±0.016         | 0.695±0.039        | 0.652±0.022 | 0.159±0.009        | 0.279±0.018        | 0.236±0.013        | 0.683±0.019        | 0.494±0.124        | 0.6±0.032          | <b>0.225±0.034</b> | 0.364±0.034        | 0.308±0.054       |
| Cora    | Degree     | VS             | <b>0.829±0.012</b> | 0.73±0.043         | 0.693±0.051 | 0.166±0.011        | 0.264±0.009        | 0.197±0.033        | <b>0.717±0.019</b> | 0.429±0.083        | 0.607±0.042        | 0.242±0.019        | 0.478±0.042        | 0.312±0.014       |
|         |            | Dirichlet      | 0.534±0.007        | 0.515±0.003        | 0.442±0.012 | 0.12±0.004         | 0.304±0.01         | 0.315±0.004        | 0.414±0.007        | 0.507±0.004        | 0.419±0.006        | 0.163±0.002        | 0.28±0.006         | 0.338±0.004       |
|         |            | ETS            | <b>0.581±0.003</b> | 0.576±0.011        | 0.516±0.013 | 0.301±0.009        | 0.317±0.018        | 0.285±0.007        | 0.47±0.002         | <b>0.563±0.003</b> | 0.496±0.005        | 0.31±0.077         | 0.373±0.009        | 0.311±0.006       |
|         |            | IRM            | <b>0.582±0.002</b> | 0.579±0.009        | 0.523±0.008 | 0.125±0.001        | <b>0.076±0.004</b> | 0.129±0.004        | 0.469±0.001        | 0.562±0.004        | 0.494±0.004        | 0.194±0.005        | <b>0.088±0.011</b> | 0.098±0.003       |
|         |            | Orderinvariant | <b>0.581±0.003</b> | 0.562±0.004        | 0.562±0.006 | 0.226±0.024        | 0.134±0.023        | 0.126±0.012        | 0.47±0.002         | 0.561±0.004        | 0.496±0.004        | 0.154±0.022        | 0.087±0.006        | 0.076±0.006       |
| Cora    | Word       | Spline         | 0.571±0.003        | 0.58±0.006         | 0.518±0.011 | <b>0.080±0.004</b> | 0.093±0.007        | 0.092±0.007        | 0.459±0.003        | <b>0.565±0.004</b> | 0.496±0.005        | 0.158±0.01         | <b>0.091±0.009</b> | 0.128±0.012       |
|         |            | VS             | <b>0.581±0.003</b> | <b>0.581±0.005</b> | 0.529±0.005 | 0.306±0.004        | 0.313±0.006        | 0.294±0.004        | 0.47±0.001         | 0.562±0.005        | 0.498±0.008        | 0.345±0.005        | 0.368±0.016        | 0.308±0.003       |
|         |            | Dirichlet      | 0.579±0.007        | 0.575±0.004        | 0.491±0.013 | 0.105±0.011        | 0.28±0.007         | 0.282±0.012        | 0.562±0.007        | 0.586±0.009        | 0.507±0.006        | 0.095±0.006        | 0.264±0.022        | 0.249±0.007       |
|         |            | ETS            | 0.607±0.002        | <b>0.636±0.003</b> | 0.562±0.006 | 0.282±0.002        | 0.359±0.02         | 0.311±0.006        | 0.603±0.004        | 0.633±0.003        | 0.561±0.005        | 0.243±0.023        | 0.377±0.023        | 0.365±0.005       |
|         |            | IRM            | 0.608±0.001        | <b>0.632±0.004</b> | 0.562±0.006 | 0.115±0.002        | 0.124±0.006        | 0.16±0.005         | 0.602±0.003        | 0.635±0.004        | 0.557±0.006        | 0.106±0.002        | 0.136±0.012        | 0.176±0.007       |
| WebKB   | University | Orderinvariant | 0.607±0.002        | <b>0.636±0.003</b> | 0.561±0.006 | 0.174±0.024        | 0.111±0.008        | 0.095±0.006        | 0.603±0.004        | <b>0.638±0.004</b> | 0.56±0.004         | 0.154±0.022        | 0.087±0.006        | 0.076±0.006       |
|         |            | Spline         | 0.598±0.005        | 0.633±0.004        | 0.561±0.007 | <b>0.073±0.002</b> | 0.077±0.005        | <b>0.069±0.007</b> | 0.591±0.002        | 0.632±0.002        | 0.56±0.006         | <b>0.063±0.006</b> | <b>0.066±0.005</b> | 0.08±0.004        |
|         |            | VS             | 0.607±0.001        | 0.633±0.006        | 0.574±0.007 | 0.283±0.003        | 0.368±0.009        | 0.32±0.005         | 0.603±0.004        | <b>0.637±0.004</b> | 0.573±0.008        | 0.261±0.005        | 0.396±0.028        | 0.373±0.006       |
|         |            | Dirichlet      | 0.229±0.018        | 0.231±0.015        | 0.234±0.007 | 0.472±0.06         | 0.562±0.014        | 0.534±0.022        | <b>0.244±0.105</b> | 0.242±0.166        | <b>0.298±0.077</b> | <b>0.299±0.092</b> | 0.468±0.092        | 0.483±0.055       |
|         |            | ETS            | <b>0.253±0.005</b> | <b>0.277±0.007</b> | 0.234±0.003 | <b>0.64±0.06</b>   | 0.421±0.017        | 0.327±0.015        | 0.121±0.021        | 0.128±0.017        | 0.101±0.033        | 0.539±0.112        | 0.437±0.032        | <b>0.293±0.01</b> |

Table 15: **Alternative Anchoring Strategies.** Here, we consider an alternative anchoring formulation for graph classification. Namely, instead of shuffling features across the batch (denoted Batch in the table), we perform READOUT anchoring by fitting a normal distribution over the hidden representations. We then randomly sample from this distribution to create anchors. Conceptually, this is similar to the node feature anchoring strategy. One potential direction of future work that is permitted by this formulation is to optimize the parameters of this distribution given a signal from an appropriate auxiliary task or loss. For example, we could perform an alternating optimization where the GNN is trained to minimize the loss, and the mean and variance of the anchoring distribution are optimized to minimize the expected calibration error on a separate calibration dataset. While a rigorous formulation is left to future work, we emphasize that the potential for improving the anchoring distribution, and thus controlling corresponding hypothesis diversity, is in fact a unique benefit of G- $\Delta$ UQ.

| Shift Type                | Method         | Test Acc                              |                     |                                       | Test Cal                              |                                      |                                       | OOD Acc                              |                                      |                                       | OOD Cal                              |                                       |                                       |
|---------------------------|----------------|---------------------------------------|---------------------|---------------------------------------|---------------------------------------|--------------------------------------|---------------------------------------|--------------------------------------|--------------------------------------|---------------------------------------|--------------------------------------|---------------------------------------|---------------------------------------|
|                           |                | MPNN                                  | Batch               | Random                                | MPNN                                  | Batch                                | Random                                | MPNN                                 | Batch                                | Random                                | MPNN                                 | Batch                                 | Random                                |
| GoodShift_basis_concept   | Dirichlet      | 0.995 $\pm$ 0.0007                    | 0.994 $\pm$ 0.0002  | 0.996 $\pm$ 0.0009                    | 0.040 $\pm$ 0.0037                    | 0.036 $\pm$ 0.0016                   | 0.035 $\pm$ 0.0058                    | 0.924 $\pm$ 0.0069                   | 0.923 $\pm$ 0.0117                   | <b>0.942 <math>\pm</math> 0.0034</b>  | 0.080 $\pm$ 0.0153                   | 0.102 $\pm$ 0.0071                    | <b>0.063 <math>\pm</math> 0.0086</b>  |
|                           | ETS            | 0.995 $\pm$ 0.0007                    | 0.995 $\pm$ 0.0005  | 0.996 $\pm$ 0.0007                    | 0.035 $\pm$ 0.0034                    | 0.036 $\pm$ 0.0101                   | 0.032 $\pm$ 0.0052                    | 0.925 $\pm$ 0.0095                   | 0.926 $\pm$ 0.009                    | <b>0.935 <math>\pm</math> 0.0068</b>  | 0.095 $\pm$ 0.0098                   | 0.096 $\pm$ 0.0128                    | 0.087 $\pm$ 0.01451                   |
|                           | IRM            | 0.9954 $\pm$ 0.0007                   | 0.9957 $\pm$ 0.0009 | 0.9965 $\pm$ 0.0004                   | <b>0.0496 <math>\pm</math> 0.0089</b> | 0.0229 $\pm$ 0.0105                  | <b>0.0225 <math>\pm</math> 0.0038</b> | 0.9251 $\pm$ 0.0096                  | 0.9301 $\pm$ 0.0123                  | <b>0.9462 <math>\pm</math> 0.0024</b> | 0.0873 $\pm$ 0.0176                  | 0.0966 $\pm$ 0.0103                   | 0.0907 $\pm$ 0.0276                   |
|                           | OrderInvariant | 0.995 $\pm$ 0.0007                    | 0.995 $\pm$ 0.0005  | 0.995 $\pm$ 0.0005                    | 0.033 $\pm$ 0.0094                    | 0.028 $\pm$ 0.0047                   | <b>0.032 <math>\pm</math> 0.0009</b>  | 0.925 $\pm$ 0.0095                   | 0.928 $\pm$ 0.0104                   | 0.935 $\pm$ 0.0027                    | 0.090 $\pm$ 0.0092                   | 0.093 $\pm$ 0.0070                    | <b>0.0754 <math>\pm</math> 0.0029</b> |
|                           | Spline         | 0.995 $\pm$ 0.0007                    | 0.995 $\pm$ 0.0007  | <b>0.9962 <math>\pm</math> 0.0005</b> | 0.034 $\pm$ 0.0002                    | 0.035 $\pm$ 0.0090                   | 0.032 $\pm$ 0.0048                    | 0.924 $\pm$ 0.0098                   | 0.926 $\pm$ 0.0092                   | 0.937 $\pm$ 0.0030                    | 0.091 $\pm$ 0.0084                   | 0.089 $\pm$ 0.0123                    | <b>0.083 <math>\pm</math> 0.0065</b>  |
|                           | VS             | 0.995 $\pm$ 0.0007                    | 0.995 $\pm$ 0.0005  | <b>0.996 <math>\pm</math> 0.000</b>   | 0.035 $\pm$ 0.0034                    | 0.036 $\pm$ 0.0087                   | 0.033 $\pm$ 0.0098                    | 0.925 $\pm$ 0.0094                   | 0.928 $\pm$ 0.0095                   | 0.936 $\pm$ 0.0053                    | 0.094 $\pm$ 0.0066                   | 0.095 $\pm$ 0.0133                    | 0.082 $\pm$ 0.009                     |
| GoodShift_basis_covariate | Dirichlet      | 0.999 $\pm$ 0.0003                    | 0.999 $\pm$ 0.0004  | 0.999 $\pm$ 0.0002                    | 0.017 $\pm$ 0.0054                    | 0.017 $\pm$ 0.0019                   | 0.014 $\pm$ 0.0004                    | 0.685 $\pm$ 0.0504                   | 0.650 $\pm$ 0.0450                   | <b>0.698 <math>\pm</math> 0.0138</b>  | 0.336 $\pm$ 0.0667                   | 0.371 $\pm$ 0.0474                    | 0.320 $\pm$ 0.0140                    |
|                           | ETS            | <b>0.9997 <math>\pm</math> 0.0004</b> | 0.999 $\pm$ 0.0005  | 0.999 $\pm$ 0.0002                    | <b>0.0095 <math>\pm</math> 0.0091</b> | 0.017 $\pm$ 0.0064                   | 0.017 $\pm$ 0.0056                    | 0.690 $\pm$ 0.0434                   | 0.649 $\pm$ 0.0476                   | 0.686 $\pm$ 0.0226                    | 0.313 $\pm$ 0.0413                   | <b>0.3739 <math>\pm</math> 0.0485</b> | 0.334 $\pm$ 0.0167                    |
|                           | IRM            | <b>0.9997 <math>\pm</math> 0.0004</b> | 0.999 $\pm$ 0.0006  | 0.999 $\pm$ 0.0003                    | <b>0.0087 <math>\pm</math> 0.0072</b> | 0.010 $\pm$ 0.0032                   | 0.014 $\pm$ 0.0042                    | 0.690 $\pm$ 0.0434                   | 0.647 $\pm$ 0.0472                   | 0.692 $\pm$ 0.0226                    | 0.315 $\pm$ 0.0505                   | 0.354 $\pm$ 0.0450                    | 0.328 $\pm$ 0.0211                    |
|                           | OrderInvariant | <b>0.9997 <math>\pm</math> 0.0004</b> | 0.999 $\pm$ 0.0005  | 0.999 $\pm$ 0.0003                    | 0.014 $\pm$ 0.0028                    | 0.020 $\pm$ 0.0090                   | 0.013 $\pm$ 0.0081                    | 0.690 $\pm$ 0.0434                   | 0.649 $\pm$ 0.0450                   | 0.689 $\pm$ 0.0170                    | 0.320 $\pm$ 0.0501                   | 0.358 $\pm$ 0.0410                    | 0.328 $\pm$ 0.0218                    |
|                           | Spline         | <b>0.9997 <math>\pm</math> 0.0004</b> | 0.999 $\pm$ 0.0005  | 0.999 $\pm$ 0.0003                    | 0.016 $\pm$ 0.0049                    | 0.017 $\pm$ 0.0053                   | 0.017 $\pm$ 0.0052                    | 0.690 $\pm$ 0.0434                   | 0.649 $\pm$ 0.0476                   | <b>0.6923 <math>\pm</math> 0.0199</b> | 0.324 $\pm$ 0.0548                   | <b>0.3733 <math>\pm</math> 0.0507</b> | 0.327 $\pm$ 0.0105                    |
|                           | VS             | <b>0.9997 <math>\pm</math> 0.0003</b> | 0.999 $\pm$ 0.0003  | 0.999 $\pm$ 0.0002                    | 0.011 $\pm$ 0.0053                    | 0.014 $\pm$ 0.0034                   | 0.012 $\pm$ 0.0016                    | 0.682 $\pm$ 0.0561                   | 0.650 $\pm$ 0.0546                   | <b>0.687 <math>\pm</math> 0.0251</b>  | 0.325 $\pm$ 0.0568                   | <b>0.374 <math>\pm</math> 0.0591</b>  | 0.337 $\pm$ 0.0264                    |
| GOODSST2_length_concept   | Dirichlet      | 0.938 $\pm$ 0.0019                    | 0.939 $\pm$ 0.0056  | <b>0.942 <math>\pm</math> 0.0018</b>  | 0.189 $\pm$ 0.01989                   | <b>0.148 <math>\pm</math> 0.0178</b> | <b>0.187 <math>\pm</math> 0.0256</b>  | <b>0.694 <math>\pm</math> 0.0193</b> | 0.693 $\pm$ 0.0020                   | 0.687 $\pm$ 0.0027                    | <b>0.146 <math>\pm</math> 0.0196</b> | <b>0.133 <math>\pm</math> 0.015</b>   | 0.169 $\pm$ 0.0168                    |
|                           | ETS            | 0.938 $\pm$ 0.0020                    | 0.939 $\pm$ 0.0060  | 0.941 $\pm$ 0.0017                    | 0.389 $\pm$ 0.0018                    | 0.390 $\pm$ 0.0022                   | 0.393 $\pm$ 0.0007                    | <b>0.694 <math>\pm</math> 0.0193</b> | 0.692 $\pm$ 0.0019                   | 0.687 $\pm$ 0.0034                    | 0.214 $\pm$ 0.0098                   | 0.216 $\pm$ 0.0033                    | 0.220 $\pm$ 0.0057                    |
|                           | IRM            | 0.939 $\pm$ 0.0016                    | 0.939 $\pm$ 0.0058  | 0.941 $\pm$ 0.0018                    | 0.326 $\pm$ 0.0011                    | 0.326 $\pm$ 0.0013                   | 0.327 $\pm$ 0.0017                    | <b>0.697 <math>\pm</math> 0.0185</b> | 0.692 $\pm$ 0.0026                   | 0.685 $\pm$ 0.0026                    | 0.240 $\pm$ 0.0017                   | 0.232 $\pm$ 0.0050                    | 0.242 $\pm$ 0.0053                    |
|                           | OrderInvariant | 0.938 $\pm$ 0.0020                    | 0.939 $\pm$ 0.0060  | 0.941 $\pm$ 0.0022                    | 0.314 $\pm$ 0.0030                    | 0.315 $\pm$ 0.0029                   | 0.315 $\pm$ 0.0012                    | <b>0.694 <math>\pm</math> 0.0193</b> | 0.692 $\pm$ 0.0019                   | 0.687 $\pm$ 0.0033                    | 0.224 $\pm$ 0.0010                   | 0.222 $\pm$ 0.0030                    | 0.225 $\pm$ 0.0054                    |
|                           | Spline         | 0.938 $\pm$ 0.0026                    | 0.938 $\pm$ 0.0044  | 0.941 $\pm$ 0.0010                    | 0.329 $\pm$ 0.0021                    | 0.329 $\pm$ 0.0019                   | 0.328 $\pm$ 0.0012                    | <b>0.692 <math>\pm</math> 0.0190</b> | 0.692 $\pm$ 0.0022                   | 0.687 $\pm$ 0.0035                    | 0.234 $\pm$ 0.0052                   | 0.231 $\pm$ 0.0044                    | 0.243 $\pm$ 0.0034                    |
|                           | VS             | 0.938 $\pm$ 0.0027                    | 0.939 $\pm$ 0.0057  | <b>0.941 <math>\pm</math> 0.0018</b>  | 0.290 $\pm$ 0.0099                    | 0.484 $\pm$ 0.0008                   | 0.487 $\pm$ 0.0007                    | 0.693 $\pm$ 0.0184                   | 0.693 $\pm$ 0.0018                   | 0.687 $\pm$ 0.0031                    | 0.331 $\pm$ 0.0484                   | 0.375 $\pm$ 0.0022                    | 0.382 $\pm$ 0.0048                    |
| GOODSST2_length_covariate | Dirichlet      | <b>0.896 <math>\pm</math> 0.0029</b>  | 0.893 $\pm$ 0.0009  | 0.895 $\pm$ 0.00095                   | 0.196 $\pm$ 0.0155                    | <b>0.172 <math>\pm</math> 0.0091</b> | <b>0.1797 <math>\pm</math> 0.0109</b> | 0.825 $\pm$ 0.0037                   | 0.827 $\pm$ 0.0066                   | 0.805 $\pm$ 0.0150                    | 0.163 $\pm$ 0.0198                   | <b>0.141 <math>\pm</math> 0.0087</b>  | <b>0.142 <math>\pm</math> 0.0122</b>  |
|                           | ETS            | <b>0.896 <math>\pm</math> 0.0023</b>  | 0.894 $\pm$ 0.0011  | 0.894 $\pm$ 0.0006                    | 0.357 $\pm$ 0.0013                    | 0.359 $\pm$ 0.0004                   | 0.362 $\pm$ 0.0019                    | 0.826 $\pm$ 0.0036                   | <b>0.828 <math>\pm</math> 0.0065</b> | 0.806 $\pm$ 0.0117                    | 0.309 $\pm$ 0.0050                   | 0.314 $\pm$ 0.0076                    | 0.300 $\pm$ 0.0070                    |
|                           | IRM            | 0.895 $\pm$ 0.0019                    | 0.893 $\pm$ 0.0003  | 0.894 $\pm$ 0.0007                    | 0.307 $\pm$ 0.0004                    | 0.307 $\pm$ 0.0003                   | 0.306 $\pm$ 0.0020                    | 0.826 $\pm$ 0.0040                   | <b>0.828 <math>\pm</math> 0.0065</b> | 0.809 $\pm$ 0.0152                    | 0.276 $\pm$ 0.0046                   | 0.277 $\pm$ 0.0061                    | 0.265 $\pm$ 0.0078                    |
|                           | OrderInvariant | <b>0.896 <math>\pm</math> 0.0023</b>  | 0.894 $\pm$ 0.0011  | 0.894 $\pm$ 0.0008                    | 0.288 $\pm$ 0.0008                    | 0.285 $\pm$ 0.0008                   | 0.284 $\pm$ 0.0013                    | 0.826 $\pm$ 0.0036                   | <b>0.828 <math>\pm</math> 0.0065</b> | 0.806 $\pm$ 0.0106                    | 0.244 $\pm$ 0.0022                   | 0.241 $\pm$ 0.0037                    | 0.225 $\pm$ 0.0054                    |
|                           | Spline         | 0.894 $\pm$ 0.0016                    | 0.890 $\pm$ 0.0009  | 0.892 $\pm$ 0.0040                    | 0.309 $\pm$ 0.0024                    | 0.307 $\pm$ 0.0009                   | 0.307 $\pm$ 0.0022                    | 0.822 $\pm$ 0.0026                   | <b>0.822 <math>\pm</math> 0.0092</b> | 0.801 $\pm$ 0.0110                    | 0.275 $\pm$ 0.0043                   | 0.276 $\pm$ 0.0063                    | 0.264 $\pm$ 0.0063                    |
|                           | VS             | <b>0.8963 <math>\pm</math> 0.0028</b> | 0.893 $\pm$ 0.0008  | 0.894 $\pm$ 0.0007                    | 0.291 $\pm$ 0.0033                    | 0.460 $\pm$ 0.0011                   | 0.465 $\pm$ 0.0010                    | 0.821 $\pm$ 0.0053                   | 0.827 $\pm$ 0.0071                   | 0.806 $\pm$ 0.0119                    | 0.299 $\pm$ 0.01895                  | 0.433 $\pm$ 0.0061                    | 0.429 $\pm$ 0.0054                    |